DepFormer: A Unified Framework with Bimodal Collaborative Transformer for Depression Detection

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Abstract

Depression is an increasingly prevalent mental health issue worldwide, especially among the elderly, where effective early detection is crucial for timely intervention. While recent multimodal approaches demonstrate promise in leveraging visual and auditory cues for automatic depression recognition, existing methods often fail to extract fine-grained, depression-specific patterns from pre-extracted features and underutilize cross-modal interactions. To address these challenges, we propose DepFormer, a unified framework that incorporates a Bimodal Collaborative Transformer (BCT) for cross-modal representation learning and a personalized fusion module to enhance individual-specific modeling. The architecture comprises: (1) unimodal feature extraction, (2) bimodal collaborative representation learning via the BCT, (3) personalized feature fusion, and (4) final depression classification. Critically, the BCT employs symmetric bidirectional branches comprising an Audio-to-Video Transformer and a Video-to-Audio Transformer to enable mutual enhancement and complementary learning across modalities. Extensive experiments validate DepFormer's effectiveness, securing first place in the MPDD Challenge 2025 (Elderly Track), underscoring its strong practical potential.

CCS Concepts

• **Computing methodologies → Artificial intelligence**.

Keywords

Affective Computing, Automatic Depression Detection, Multimodal Fusion

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1 Introduction

Depression is a common and disabling mental disorder marked by persistent sadness, loss of interest, and cognitive decline. According to the World Health Organization, over 280 million people worldwide suffer from depression, with incidence rates continuing to rise, particularly among the elderly [1]. If not properly treated, depression can lead to severe outcomes such as social withdrawal, reduced quality of life, and even suicide. These risks underscore the need for early and reliable detection. Given that traditional clinical interviews for depression are costly and unscalable, particularly in the face of a growing patient population, the development of an automated detection system has become an urgent need [1].

Automated depression detection based on multimodal data (e.g., video, audio, and text) has shown great promise. Existing research mainly focused on two key aspects: extracting depression-relevant features from each modality and effectively fusing these heterogeneous features into a unified representation. Early works employed CNNs and RNNs to capture spatial and temporal patterns [2–8], while recent studies leverage Transformer-based architectures and state-space models like Mamba for improved long-range temporal modeling [9–15]. Regarding fusion, methods vary from decision-level aggregation of unimodal predictions [16–18] to feature-level concatenation or projection before classification [16–18]. More recent approaches explore model-level fusion with cross-modal attention mechanisms to enable richer interaction [10, 15, 19, 20].

While these methods have achieved promising results, they still face two key limitations: (1) insufficient extraction of depression-relevant patterns from pre-extracted features (e.g., MFCCs, OpenFace), which often suffer from semantic loss and limited expressiveness due to the unavailability of raw data under privacy constraints; and (2) sub-optimal multimodal fusion strategies that fail to fully

[1] https://www.who.int/news-room/fact-sheets/detail/depression

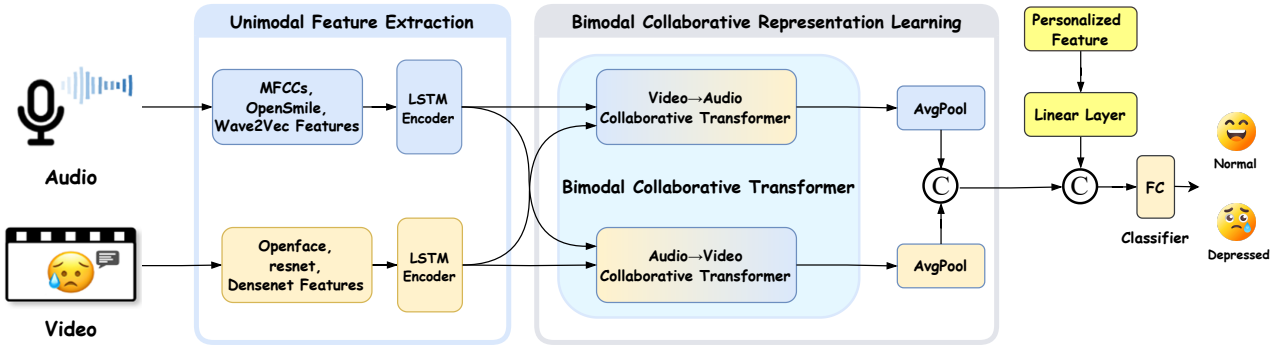

Figure 1: Overview of DepFormer. Temporal features from pre-extracted audio and video inputs are refined by a Bimodal Collaborative Transformer with personalized information, then used for depression classification.

exploit the complementary nature of different modalities. To address these challenges, we propose DepFormer, a unified framework that leverages a Bimodal Collaborative Transformer for cross-modal representation learning, along with personalized feature fusion to enhance depression detection. As illustrated in Fig. 1, the model begins with unimodal feature extraction modules that learn modality-specific temporal representations from pre-extracted feature sequences. The extracted representations are then fed into a Bimodal Collaborative Transformer, which enables bidirectional interaction between audio and video features, allowing each modality to enrich the other through a cross-attention mechanism. To further enhance adaptability, we incorporate personalized fusion, aligning the learned representation with individual-specific traits to better reflect subject variability. Finally, the joint representation, which is enriched through cross-modal interaction and personalized adaptation, is used for depression classification. Extensive experimental results on benchmark dataset demonstrate that our method achieves superior performance for depression detection.

Our main contributions can be summarized as follows:

- We propose a unified framework for multimodal depression detection that integrates Bimodal Collaborative Transformers and personalized fusion to model temporal dependencies and cross-modal interactions effectively.
- We design a dual-branch collaborative transformer structure that facilitates modality-specific enhancement through mutual guidance and complementary learning.
- Extensive experiments demonstrate the effectiveness of our method, and we achieved rank 1 in the MPDD 2025 [9] Elderly Track, highlighting the practical value and robustness of our approach.

2 Related Work

2.1 Multimodal Depression Detection

Automatic depression detection has attracted increasing attention in recent years, especially with the availability of multimodal behavioral data such as audio and video. However, unlike many other multimodal tasks, depression detection often involves sensitive information, making the collection and release of raw audio and video data difficult due to privacy concerns. As a result, datasets such as DAIC-WOZ [21], D-Vlog [10] and MPDD [9] provide only pre-extracted features instead of raw video and audio, which brings new challenges to depression detection.

Early studies on multimodal depression detection relied on handcrafted features. Visual cues were captured using local descriptors like LBP and LBP-TOP [22, 23], while MFCCs [24] and LIWC [25] were commonly used for audio and text, respectively. Although interpretable and easy to implement, these methods depend heavily on manually designed features, limiting their capacity to model complex patterns and generalize to real-world scenarios.

Recent advances in deep learning enable richer representation learning from multimodal features. To model temporal visual dynamics, CNN-based methods have been widely explored, typically using 3D CNNs or 2D CNNs combined with temporal modules such as attentive convolutions [2–5]. However, these methods are limited in capturing long-range dependencies. To address this, RNN-based models, particularly those using LSTM units, have been introduced to better capture temporal progression in visual and acoustic cues. These methods [6–8] leverage sequential modeling to capture dynamic changes in mood-related cues. While such models enhance sequential modeling, their generalization to real-world depression analysis remains limited, especially under diverse subject conditions and varying recording environments [26].

More recently, Transformer-based architectures [9–14] and Mamba-based models [15] have been applied to depression detection, offering enhanced long-range dependency modeling through attention or state-space mechanisms. For example, Yoon et al. [10] proposed a Transformer-based multimodal model with cross-attention to effectively capture relationships across acoustic and visual features, while Ye et al. [15] introduced DepMamba, leveraging Mamba for hierarchical contextual modeling and progressive fusion. Building on these developments, we introduce a unified framework incorporating a Bimodal Collaborative Transformer for bimodal representation learning and personalized depression severity classification.

2.2 Multimodal Fusion Methods

In recent years, multimodal fusion techniques, which integrate cues from heterogeneous sources such as audio and video, have become important for improving depression detection. These multimodal

fusion approaches can typically be categorized into feature-level, decision-level, and model-level fusion strategies.

Feature-level fusion involves concatenating features from various modalities to learn a unified representation before classification [16–18]. For instance, Fu et al. [16] fused facial action units and speech features by extracting complementary emotional features by integrating them via a graph-based temporal attention model to enhance depression detection. However, feature-level fusion may suffer from feature incompatibility and high dimensionality when combining heterogeneous modalities.

Alternatively, decision-level fusion ensembles the predictions from models trained on single modalities to reach a collective classification [27–30]. Zhang et al. [28] proposed a decision-level method using a multi-agent system. While straightforward, decision-level fusion lacks deep interaction between modalities, potentially limiting its ability to capture subtle cross-modal correlations.

Model-level fusion incorporates cross-modal interactions within the model architecture, allowing more integrated learning [10, 15, 19, 20]. Tao et al. [20] proposed DepMSTAT, a model-level fusion framework that jointly captures spatio-temporal acoustic and visual features with a unified transformer for depression detection. In contrast, our work explores a different direction by leveraging a bimodal collaborative transformer to dynamically enhance modality-specific representations through mutual refinement, further incorporating personalized features for robust classification.

3 Methodology

In this section, we present our proposed DepFormer, a unified multimodal depression detection framework, which integrates acoustic, video, and personalized features to achieve effective depression detection. The core idea of our model is to first extract modality-specific temporal representations, then enhance them through a Bimodal Collaborative Transformer, and finally incorporate personalized information to form a unified representation for final classification. The entire architecture consists of four key stages: Unimodal Feature Extraction, Bimodal Collaborative Representation Learning, Personalized Feature Fusion, and Depression Classification. An overview of the model is illustrated in Fig. 1.

3.1 Unimodal Feature Extraction

In the first stage of our model, we encode acoustic and video temporal sequences independently. Given the inherently sequential nature of both audio and video data, we adopt Long Short-Term Memory (LSTM) networks to capture temporal dependencies within each modality. This allows the model to learn modality-specific representations that preserve important temporal dynamics.

Let the acoustic and video feature sequences be denoted as $\mathbf{X}_a \in \mathbb{R}^{T_a \times D_a}$ and $\mathbf{X}_v \in \mathbb{R}^{T_v \times D_v}$, where T_a and T_v represent the sequence lengths, and D_a and D_v are the feature dimensions of the acoustic and video modalities, respectively.

We employ two separate LSTM encoders to model each modality's temporal structure, with outputs computed as:

$$\mathbf{L}_a = \mathrm{LSTM}_{\mathrm{audio}}(\mathbf{X}_a), \quad \mathbf{L}_v = \mathrm{LSTM}_{\mathrm{video}}(\mathbf{X}_v). \tag{1}$$

Each LSTM encoder produces a sequence of high-level representations, followed by a dense transformation layer applied at every timestep:

$$\mathbf{F}_a = \tanh(\mathbf{L}_a \mathbf{W}_a), \quad \mathbf{F}_v = \tanh(\mathbf{L}_v \mathbf{W}_v), \tag{2}$$

where $\mathbf{W}_a \in \mathbb{R}^{D_a \times d_{\mathrm{model}}}$, $\mathbf{W}_v \in \mathbb{R}^{D_v \times d_{\mathrm{model}}}$ are learnable projection matrices specific to each modality. This dense embedding strategy enables the model to transform raw LSTM outputs into a compact, non-linear feature space while preserving temporal resolution. The resulting modality-specific embeddings $\mathbf{F}_a$ and $\mathbf{F}_v$ are then forwarded to the Bimodal Collaborative Representation Learning module.

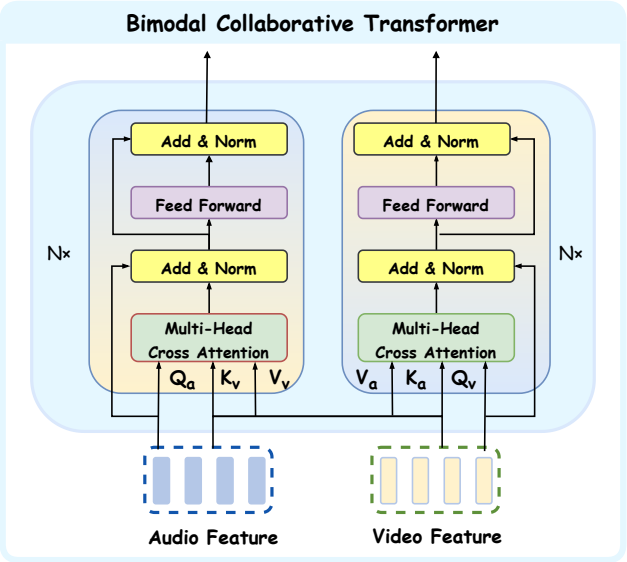

Figure 2: Illustration of Bimodal Collaborative Transformer.

3.2 Bimodal Collaborative Representation Learning

A key component of our model is the **Bimodal Collaborative Transformer** (BCT), which is designed to enable deep interaction between the acoustic and video modalities. As illustrated in Fig. 2, this module consists of two symmetric branches: an *Audio-to-Video Collaborative Transformer* and a *Video-to-Audio Collaborative Transformer*. Each branch is implemented as a stack of N Transformer layers, where each layer refines the temporal representation of one modality by attending to the complementary modality through multi-head cross-attention.

Each layer in the collaborative Transformer branch adopts a standard Transformer architecture, where self-attention is replaced by multi-head cross-attention to capture inter-modal dependencies. The layer is composed of a cross-attention module, residual connections, layer normalization, and a feed-forward network.

Let $M^v_{\mathrm{attn}} \in \{0, -\infty\}^{T_v \times T_v}$ and $M^a_{\mathrm{attn}} \in \{0, -\infty\}^{T_a \times T_a}$ denote the attention masks for the visual and acoustic modalities, respectively. Let $\mathbf{F}_a^{[i]}$ and $\mathbf{F}_v^{[i]}$ denote the input acoustic and visual features to the i-th Transformer layer, where $i \in \{1, \ldots, N\}$. The inputs to the first layer are initialized as $\mathbf{F}_a^{[0]} = \mathbf{F}_a$ and $\mathbf{F}_v^{[0]} = \mathbf{F}_v$.

The layer-wise computation for the Video-to-Audio branch is detailed below, with the Audio-to-Video branch following the same structure but with modality roles reversed:

(1) Multi-Head Cross-Attention. We first project the input features into query, key, and value representations:

$$\mathbf{Q}_a^{[i]} = \mathbf{F}_a^{[i]}\mathbf{W}_Q, \quad \mathbf{K}_v^{[i]} = \mathbf{F}_v^{[i]}\mathbf{W}_K, \quad \mathbf{V}_v^{[i]} = \mathbf{F}_v^{[i]}\mathbf{W}_V, \tag{3}$$

where $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{d_{\text{model}} \times (hd_{\text{model}})}$ are projection matrices. The projected features are reshaped and split into h heads, each of dimension d_{model}. For the i-th layer and j-th attention head, the query matrix is $\mathbf{Q}_{a,j}^{[i]} \in \mathbb{R}^{T_a \times d_{\text{model}}}$, and the key and value matrices are $\mathbf{K}_{v,j}^{[i]}, \mathbf{V}_{v,j}^{[i]} \in \mathbb{R}^{T_v \times d_{\text{model}}}$. The attention output for the j-th head is computed as:

$$\text{head}_j^{[i]} = \text{softmax}\left(\frac{\mathbf{Q}_{a,j}^{[i]}(\mathbf{K}_{v,j}^{[i]})^\top + \mathbf{M}_v^{\text{attn}}}{\sqrt{d_{\text{model}}}}\right)\mathbf{V}_{v,j}^{[i]}, \tag{4}$$

The outputs from all heads are concatenated and projected to form the final result:

$$\text{MultiHead}^{[i]} = \text{Concat}(\text{head}_1^{[i]}, \ldots, \text{head}_h^{[i]})\mathbf{W}^O, \tag{5}$$

where $\mathbf{W}^O \in \mathbb{R}^{(hd_{\text{model}}) \times d_{\text{model}}}$ is the output projection matrix.

(2) Add & LayerNorm. A residual connection and layer normalization are applied:

$$\mathbf{Z}_a^{[i]} = \text{LayerNorm}(\mathbf{F}_a^{[i]} + \text{MultiHead}(\mathbf{F}_a^{[i]}, \mathbf{F}_v^{[i]}, \mathbf{M}_v^{\text{attn}})). \tag{6}$$

(3) Feed-Forward & Normalization. Each block concludes with a feed-forward network, followed by a residual connection and layer normalization:

$$\mathbf{F}_a^{[i+1]} = \text{LayerNorm}(\mathbf{Z}_a^{[i]} + \text{FFN}(\mathbf{Z}_a^{[i]})), \tag{7}$$

where $\text{FFN}(\cdot)$ denotes a two-layer fully connected network with ReLU activation.

To obtain fixed-length representations, we apply masked average pooling over the sequence length dimension of the final-layer outputs from both branches. Specifically, $\mathbf{F}_a^{[N]}$ denotes the final output of the Video-to-Audio branch, and $\mathbf{F}_v^{[N]}$ corresponds to the final output of the Audio-to-Video branch. The resulting global representations for each modality can be formulated as:

$$\begin{aligned} \hat{\mathbf{F}}_a &= \text{MaskedAvgPool}(\mathbf{F}_a^{[N]}, \mathbf{M}_a^{\text{pool}}), \\ \hat{\mathbf{F}}_v &= \text{MaskedAvgPool}(\mathbf{F}_v^{[N]}, \mathbf{M}_v^{\text{pool}}), \end{aligned} \tag{8}$$

where $\mathbf{M}_a^{\text{pool}} \in \{0,1\}^{T_a}$ and $\mathbf{M}_v^{\text{pool}} \in \{0,1\}^{T_v}$ are binary masks used to ignore padded positions. The final bimodal representation is obtained by concatenating the modality-specific embeddings:

$$\mathbf{F}_{\text{bimodal}} = \text{Concat}(\hat{\mathbf{F}}_a, \hat{\mathbf{F}}_v). \tag{9}$$

3.3 Personalized Feature Fusion

As the MPDD dataset provides static personalized features $\mathbf{P} \in \mathbb{R}^{d_p}$, which capture participant-specific information. To integrate these features with the dynamic bimodal representations, we first project the original personalized features $\mathbf{P}$, which have dimension d_p, into the model's hidden feature space $\mathbb{R}^{d_{model}}$ via a linear layer:

$$\mathbf{P}' = \text{Linear}(\mathbf{P}), \quad \mathbf{P}' \in \mathbb{R}^{d_{\text{model}}}. \tag{10}$$

The personalized embedding $\mathbf{P}'$ is then concatenated with the bimodal collaborative feature $\mathbf{F}_{\text{bimodal}}$ to form the final feature representation:

$$\mathbf{F}_{\text{final}} = \text{Concat}(\mathbf{F}_{\text{bimodal}}, \mathbf{P}'). \tag{11}$$

This fusion strategy allows the model to leverage both the static participant-specific attributes and the dynamic audio-visual features for improved prediction.

3.4 Depression Classification

To improve robustness and mitigate class imbalance, we adopt a dual-head classification strategy. The final fused representation $\mathbf{F}_{\text{final}}$ is fed into two parallel classifiers: one trained with cross-entropy loss and the other with focal loss. Formally, the two sets of logits are given by:

$$\mathbf{z}_{\text{CE}} = \text{Classifier}_{\text{CE}}(\mathbf{F}_{\text{final}}), \quad \mathbf{z}_{\text{FL}} = \text{Classifier}_{\text{FL}}(\mathbf{F}_{\text{final}}), \tag{12}$$

where $\mathbf{z}_{\text{CE}}$ and $\mathbf{z}_{\text{FL}}$ are the logits predicted by the two classifiers.

The total training loss is defined as:

$$\mathcal{L} = \mathcal{L}_{\text{CE}}(\mathbf{z}_{\text{CE}}, \mathbf{y}) + \lambda_{\text{FL}} \cdot \mathcal{L}_{\text{Focal}}(\mathbf{z}_{\text{FL}}, \mathbf{y}), \tag{13}$$

where $\mathcal{L}_{\text{CE}}$ and $\mathcal{L}_{\text{Focal}}$ denote the cross-entropy and focal loss functions, respectively, $\mathbf{y}$ is the ground-truth label, and λ_{FL} controls the contribution of the focal loss.

Only the $\text{Classifier}_{\text{CE}}$ is used during inference, while both heads are trained jointly to improve robustness. The final prediction is obtained by applying a softmax to the logits $\mathbf{z}_{\text{CE}}$:

$$\hat{\mathbf{y}} = softmax(\mathbf{z}_{\text{CE}}). \tag{14}$$

4 Experimentation

4.1 Dataset and Evaluation Metrics

The MPDD-Elderly dataset is a multimodal dataset focused on depression detection in elderly individuals. It includes audio-visual recordings and personalized information to better capture depression-related cues while considering individual differences. Each sample is associated with all three classification schemes: binary (normal vs. depressed), ternary (normal, mild, severe), and quinary (five-level PHQ-9 severity).

For evaluation, we adopt two standard classification metrics: Accuracy (Acc) and F1 Score (F1). Both weighted and unweighted versions are calculated to evaluate performance on the MPDD-Elderly dataset.

4.2 Implementation Details

We evaluate our model on three classification tasks—binary, ternary, and quinary—under two split window settings (1-second and 5-second clips). The model is implemented in PyTorch and trained using either the Adam or AdamW optimizer, with hyperparameters summarized in Table 2.

Our model is implemented using PyTorch. All experiments are conducted on a machine equipped with an NVIDIA RTX 3090 GPU. Guided by empirical results, we set the hidden dimension d_{model} to 256, the number of Transformer layers N to 1, and the number of attention heads h to 2 to balance efficiency and performance.. For input features, we use DenseNet-based visual features for the video

Table 1: Results (%) of different methods on MPDD-Elderly. (W/U: weighted/unweighted)

Task	Window	Method	Audio	Video	Acc (W)	Acc (U)	F1 (W)	F1 (U)
Binary	1s	Baseline [9]	mfcc	openface	85.40	84.62	85.71	79.13
		DepMamba [15]	mfcc	densenet	85.72	94.27	93.98	89.15
		Ours	mfcc	densenet	**88.75**	**96.03**	**95.83**	**92.48**
	5s	Baseline [9]	opensmile	resnet	75.40	80.77	81.75	72.37
		DepMamba [15]	mfcc	densenet	73.75	90.75	89.35	79.55
		Ours	mfcc	densenet	**91.25**	**96.92**	**96.80**	**94.29**
Ternary	1s	Baseline [9]	opensmile	resnet	55.49	56.41	56.48	55.64
		DepMamba [15]	mfcc	densenet	71.88	74.32	74.23	72.85
		Ours	mfcc	densenet	**81.55**	**86.34**	**85.93**	**83.23**
	5s	Baseline [9]	wav2vec	openface	59.62	57.69	58.22	59.37
		DepMamba [15]	mfcc	densenet	72.82	69.47	69.37	68.65
		Ours	mfcc	densenet	**85.21**	**81.50**	**81.73**	**81.45**
Quinary	1s	Baseline [9]	opensmile	densenet	45.79	69.23	66.26	46.66
		DepMamba [15]	mfcc	densenet	48.87	77.09	73.10	49.55
		Ours	mfcc	densenet	**70.23**	**81.50**	**76.82**	**73.38**
	5s	Baseline [9]	mfcc	densenet	57.71	78.21	75.62	58.40
		DepMamba [15]	mfcc	densenet	61.54	78.85	75.78	55.62
		Ours	mfcc	densenet	**70.08**	**79.30**	**76.2**	**67.71**

Table 2: Training hyperparameters for each task

Task	Optimizer	Batch Size	Learning Rate	Epochs
1s Binary	AdamW	8	2e-5	300
5s Binary	AdamW	2	1.8e-5	200
1s Ternary	Adam	4	4.58e-5	200
5s Ternary	Adam	4	4.58e-5	300
1s Quinary	Adam	1	9.8e-5	300
5s Quinary	Adam	1	9.75e-5	200

modality and MFCCs for the audio modality, as these configurations yielded superior performance in our empirical evaluations. During training, input sequences within each batch are padded to the maximum sequence length in that batch, and corresponding attention masks are generated to ensure proper temporal modeling. For more details, please refer to our code[2].

Note: For the final competition submission, we adopt two strategies to further boost model performance. First, for the 1s and 5s quinary classification tasks, we perform ensemble learning by averaging predictions from multiple checkpoints, weighted by their performance. Second, to ensure label consistency across clips from the same subject, we apply a subject-level voting strategy. Specifically, since each subject may have multiple non-overlapping clips in the test set, we first predict each clip independently, then assign the majority-voted label across clips as the final subject-level prediction, which is propagated back to all associated samples.

4.3 Comparison with Existing Methods

We compare our method with the official baseline and a recent state-of-the-art model, DepMamba [15], which adopts a progressive audio-visual fusion strategy for multimodal depression detection. It leverages convolutional networks and Mamba layers for hierarchical sequence modeling and multimodal fusion. To ensure fair comparison, we incorporate the same personalized feature embeddings into the reproduced DepMamba model using our integration strategy. All methods are evaluated on the test set of the MPDD-Elderly dataset across binary, ternary, and quinary classification tasks under both 1-second and 5-second window settings.

Table 1 presents the results across binary, ternary, and quinary classification tasks under both 1-second and 5-second window settings. Our method consistently outperforms both DepMamba and the official baseline in all settings, achieving substantial gains in both accuracy and F1 scores. Notably, the performance advantage holds across all levels of label granularity, from coarse (binary) to fine-grained (quinary) classification, showing the effectiveness of our approach in recognizing varying degrees of depression severity.

4.4 Performance Fluctuation Analysis

Given the inherent variability among participants in the MPDD-Elderly dataset (e.g., health conditions, personal traits), we observe noticeable performance fluctuations across different training runs. To provide a more comprehensive evaluation, we evaluate all models on the MPDD-Elderly test set by repeating each experiment 10 times, reporting the mean and standard deviation of Accuracy and F1, and conducting paired t-tests to assess significance.

As shown in Table 3, compared to DepMamba, our method generally achieves higher average accuracy and F1 scores across all tasks. For example, in the 5s quinary classification task, our model reaches an average accuracy of 68.09% with a standard deviation of 8.26, while DepMamba achieves 55.20% with a larger fluctuation (std = 17.91). A similar pattern is observed in the F1 score (65.95 ± 7.97 vs. 50.04 ± 19.00). These improvements are not only substantial but also statistically significant in most tasks and metrics, as indicated by paired t-tests.

[2]https://github.com/lfy-ustc/MPDD25-Track1-Solution

Table 3: Mean ± SD (%) of accuracy and F1 score, averaged over weighted and unweighted results. Asterisks indicate significance over the DepMamba (paired t-test: * $p < 0.05$, ** $p < 0.01$).

Task	Method	Acc	F1
1s Binary	DepMamba [15]	81.85 ± 8.23	81.81 ± 7.48
	Ours	**81.92 ± 8.38**	**82.34 ± 7.86**
1s Ternary	DepMamba [15]	60.27 ± 8.64	60.50 ± 8.56
	Ours	**69.19± 11.49***	**68.31 ± 11.61***
1s Quinary	DepMamba [15]	59.38 ± 15.24	56.67 ± 14.96
	Ours	**66.63 ± 11.36****	**64.86 ± 9.61****
5s Binary	DepMamba [15]	77.67 ± 8.94	77.07 ± 8.18
	Ours	**85.94 ± 7.25****	**85.23 ± 7.61****
5s Ternary	DepMamba [15]	55.77 ± 12.14	54.81 ± 11.32
	Ours	**67.07 ± 12****	**65.31 ± 11.97****
5s Quinary	DepMamba [15]	55.20 ± 17.91	50.04 ± 19
	Ours	**68.09 ± 8.26****	**65.95 ± 7.97****

Despite these improvements, we still observe non-trivial variance across different training runs, especially in the fine-grained (ternary, quinary) tasks. This highlights the challenge posed by the inherent variability of the dataset. Overall, our method demonstrates relatively better consistency than DepMamba, but stability remains an open issue deserving further investigation.

4.5 Ablation Study

To investigate the contribution of each key component in our model, we conduct an ablation study on the 5-second quinary classification task using the MPDD-Elderly dataset. We choose this setting due to its higher difficulty and finer-grained classification, which provides a more rigorous evaluation of each module's effectiveness. All results are evaluated on the test set and reported as the mean and standard deviation over 5 independent runs. The results are summarized in Table 4.

Table 4: Ablation study results (wo = without the component). Metrics are averaged over weighted and unweighted results.

Methods	ACC	F1
Unimodal Feature Extraction		
wo LSTM encoder	60.20±15.83	57.47±15.59
wo Dense layer	61.93±9.88	59.38±9.00
Bimodal Feature Fusion		
wo BCT	60.20±8.19	57.53±10.35
wo Cross-attention	60.13±15.07	59.01±12.80
Personalized Feature Fusion		
wo personalized feature	64.17±10.08	61.50±9.81
ours	**69.33±8.21**	**66.91±7.59**

Unimodal Feature Extraction As shown in Table 4, replacing the LSTM encoder with a 1D convolutional layer results in a marked performance drop, indicating the effectiveness of sequential modeling for temporal features. Additionally, removing the dense projection layer (by directly applying max pooling) also leads to degraded performance, underscoring its role in transforming and enhancing the unimodal representations before fusion.

Bimodal Feature Fusion Removing the Bimodal Collaborative Transformer (BCT) entirely causes a notable reduction in both accuracy and F1 score, confirming its importance in cross-modal integration and alignment. Moreover, replacing the cross-attention mechanism within BCT with a standard self-attention module yields further performance degradation. These results highlight the necessity of explicitly modeling inter-modality interactions to enhance multimodal alignment.

Personalized Feature Fusion As shown in Table 4, removing the personalized features leads to a moderate performance drop. This suggests that while personalized information is not the dominant factor, it helps the model better adapt to individual differences, which is particularly valuable in depression detection tasks where personal context is critical.

4.6 Presentation of Final Results

As shown in Table 5, our method ranks first on the MPDD-Elderly test set with a final score of 0.8623, achieving the highest accuracy and competitive F1 score among all submissions. This result highlights the effectiveness of our approach.

Table 5: Final scores on MPDD-Elderly test datasets.

Team	Score	Accuracy	F1 Score
wakk (ours)	**0.8623**	**0.8613**	**0.8633**
ZBX_LMC	0.8580	0.8519	0.8640
SEU_AIPLab	0.8576	0.8611	0.8541
y61329697	0.8464	0.8446	0.8482

5 Conclusion

This paper presents a unified framework for multimodal depression detection, aiming to enhance the learning of depression-relevant features from pre-extracted multimodal inputs and to improve cross-modal fusion. To this end, we propose a Bimodal Collaborative Transformer that facilitates bidirectional interaction between modalities, along with a personalized fusion strategy to incorporate individual-specific information. Experimental results on the MPDD benchmark demonstrate that our method achieves state-of-the-art performance and ranks first in the MPDD-Elderly track, confirming its effectiveness and practical value for privacy-constrained, real-world applications.

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